Paper IV – Recent Advances (MCh Urology, Summer 2016)

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IV-1. Orthotopic versus Non Orthotopic bladder.

Type: LAQ (25 marks)

Read as: orthotopic neobladder versus non-orthotopic (heterotopic) urinary diversion after radical cystectomy.

Introduction and types

- **Ileal conduit** (Bricker, 1950) remains the commonest diversion worldwide; **ureterosigmoidostomy** is the oldest continent diversion.
- AUA (American Urological Association)/SUO (Society of Urologic Oncology) and EAU (European Association of Urology): counsel every patient on all options. **80–90% of men and ~75% of women** are medically eligible for a neobladder, but far fewer receive one.

Group	Examples	How urine is held and emptied
Orthotopic	Studer, Hautmann W, T pouch, Padua, Mainz III	Native rhabdosphincter; voids by relaxing and straining
Incontinent cutaneous	Ileal conduit; transverse colon, sigmoid or jejunal conduit	Stoma and appliance
Continent cutaneous	Indiana pouch; right colon pouch with appendix (Mitrofanoff) or Yang–Monti channel; Kock (historical)	Self-catheterisation through a valved channel
Rectal	Ureterosigmoidostomy, Mainz II (sigma-rectum pouch)	Anal sphincter; urine mixed with stool

Principles of a continent reservoir

- **Detubularisation** abolishes coordinated peristaltic contractions. Folding into a **sphere** gives the largest volume for a given surface area. Result: a low-pressure reservoir.
- **Volume**: 44 cm of ileum double-folded holds <200 mL at first and 500–600 mL within a year.
- **Reliable outlet**: healthy urethra and rhabdosphincter (orthotopic), or a catheterisable efferent channel (cutaneous). Continence is a balance of low reservoir pressure against outlet resistance.

Patient selection

	Orthotopic only	All continent diversions
Absolute contraindications	Tumour at the urethral margin (frozen section); significant urethral stricture	Severe renal or hepatic dysfunction; physical or mental inability to self-catheterise
Relative	Poor external sphincter; bladder-neck or anterior vaginal wall tumour; prostatic stromal invasion; irradiated sphincter	Frailty, poor cognition, social problems, prior radiation or bowel resection

- **Renal function**: creatinine <1.7–2.2 mg/dL or eGFR (estimated glomerular filtration rate) **>35–40 mL/min**, because the kidney must excrete acid reabsorbed from stored urine. Formal tests: urine pH ≤5.8 after ammonium chloride, osmolality ≥600 mOsm/kg after water deprivation.
- **Urethral recurrence**: in men the strongest predictor is prostatic stromal invasion; in women, bladder-neck or anterior vaginal wall tumour. **Intraoperative frozen section** of the urethral margin decides. CIS (carcinoma in situ) or multifocality alone do not exclude a neobladder.
- **Age and obesity are not contraindications**; comorbidity, cognition and support decide. A conduit suits the frail or unsupported patient.
- Continent cutaneous pouch: continence wanted but urethra unusable.
- Mark a conduit stoma site in everyone: up to ~10% change plan intraoperatively.

Comparison

Feature	Orthotopic neobladder	Ileal conduit	Continent cutaneous	Ureterosigmoidostomy
Appliance	None	Bag	None; small stoma	None
Body image	Best	Worst	Good	Good
Complexity	High	Lowest	Highest	Moderate
Continence	Day 80–90%; night incontinence 20–50%	Not applicable	High (Indiana, Mitrofanoff)	Depends on anal sphincter
Catheterisation	10–20% of men, ~30% of women	None	Always	None
Bowel used	Long ileal segment (Studer 54–56 cm)	10–15 cm ileum	Right colon + terminal ileum	None isolated
Typical late problems	Night leakage, retention, acidosis	Stomal stenosis (20–24%), parastomal hernia, renal deterioration	Stomal stenosis, stones, difficult catheterisation	Severe acidosis, pyelonephritis, cancer at the anastomosis

Metabolic complications (bowel in the urinary tract)

Segment	Disturbance
Ileum, colon	**Hyperchloraemic metabolic acidosis** (ammonium chloride absorbed); hypokalaemia, worse with colon
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis

Segment	Disturbance
Stomach	Hypochloraemic, hypokalaemic alkalosis

- Acidosis is severe in most ureterosigmoidostomies; most colonic (Mainz) pouches need alkali. Treat with sodium bicarbonate or citrate, and replace potassium as well.
- **Also know:** osteomalacia; vitamin B12 deficiency and bile-salt diarrhoea after ileal loss (liver B12 stores last 3–5 years); stones (ileal conduit 10–12%, continent caecal reservoirs ~20%); bacteriuria; drug reabsorption (methotrexate, phenytoin); ammoniagenic encephalopathy; growth delay in children.
- **Secondary cancer**: ureterosigmoidostomy 6–29% after 10–20 years (yearly endoscopy from 5 years); ileal conduit and ileal neobladder lowest.

Surgical complications

- **Early (≤90 days)**: ileus, urine leak (long pouch suture lines; drain percutaneously), infection; not more than with a conduit.
- **Late**: ureteroileal stricture ~3–6% (Wallace or direct end-to-side); neobladder retention, urethral stricture, rare pouch rupture; ventral and parastomal hernia.

Quality of life

- No randomised trial. Body image favours the neobladder, urinary function the conduit; no diversion is superior on HRQoL (health-related quality of life).

Recent advances

- RARC (robot-assisted radical cystectomy): **RAZOR (2018)** non-inferior 2-year progression-free survival (open diversion in both arms); **iROC**: intracorporeal diversion gave more days alive and out of hospital at 90 days.
- Intracorporeal neobladders and nerve- or organ-sparing cystectomy are increasing.

Diagram to draw: Studer neobladder and ileal conduit side by side.

Key points to remember:

- Neobladder needs eGFR >35–40, a tumour-free urethral margin and ability to self-catheterise.
- Day continence 80–90%; night incontinence 20–50%.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- No diversion is proven best for quality of life; counsel all options.

IV-2. Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS.

Type: LAQ (25 marks)

Introduction

- Stone disease is a chronic metabolic disorder; about half recur within 5–10 years. Recurrent formers need a **metabolic evaluation** and targeted therapy.
- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy inside the kidney.

Evaluation that guides therapy

- **Stone analysis** (infrared spectroscopy or X-ray diffraction); serum calcium, uric acid, creatinine, electrolytes, PTH (parathyroid hormone) if calcium is high-normal.
- **Two 24-h urines** on usual diet, at least 4 weeks after an episode: volume, calcium, oxalate, citrate, uric acid, pH, sodium, cystine.
- Targets: volume >2.5 L, calcium <250 mg (women) or <300 mg (men), oxalate <40 mg, citrate >320 mg, uric acid <600 mg.

General measures (every former)

- **Fluid** for urine output ≥2.5 L/day. **Borghi 1996**: urine >2 L cut 5-year recurrence (12.1% vs 27%).
- **Normal calcium**, low sodium (≤2.3 g/day), less animal protein, more fruit and vegetables (DASH (Dietary Approaches to Stop Hypertension)-type diet). **Borghi 2002**: this diet halved recurrence compared with a low-calcium diet. Restricting calcium raises oxalate absorption.
- Citrus juices (lemonade, orange) are adjuncts only.

Selective drug therapy

Abnormality	Treatment	Mechanism and notes
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption plus volume contraction. Add potassium citrate: hypokalaemia causes hypocitraturia
Primary hyperparathyroidism	Parathyroidectomy	Cures resorptive hypercalciuria
Hypocitraturia, distal RTA (renal tubular acidosis), chronic diarrhoea	**Potassium citrate 30–60 mEq/day**	Alkali load reduces proximal citrate reabsorption; avoid urine pH ≥7
Hyperuricosuric calcium oxalate	Allopurinol 300 mg/day	Reduce dose in renal impairment
Uric acid stones	Potassium citrate to **pH 6.0–6.5**; allopurinol only for hyperuricosuria or failed alkalinisation	A pH disease: pKa 5.5; dissolution possible
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; liquid potassium citrate; magnesium; cholestyramine	Calcium binds oxalate in the gut
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA (ribonucleic acid) interference against glycolate oxidase)	ILLUMINATE-A: 84% vs 0% reached near-normal urine oxalate

Abnormality	Treatment	Mechanism and notes
Cystinuria	Urine ≥3 L/day; low sodium and protein; potassium citrate to pH 7.0–7.5; then **tiopronin** (or D-penicillamine)	Aim cystine <250 mg/L; thiols form soluble mixed disulfides
Struvite	Complete clearance; culture-specific antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily if surgery impossible	Urease inhibitor; risk of thromboembolism

- Recurrent calcium stones with a normal work-up: thiazide or potassium citrate (AUA (American Urological Association)).
- **Trials**: Ettinger 1988 (chlorthalidone) and Barcelo 1993 (potassium citrate) reduced new stones. **NOSTONE (2023)**: hydrochlorothiazide 12.5–50 mg did not reduce recurrence, which questions the older trials.
- **Follow-up**: repeat 24-h urine within 6 months, then yearly; serum potassium 1–2 weeks after starting a thiazide or citrate; imaging for new stones.

Recent advances in RIRS instrumentation

Advance	Principle	Benefit or limitation
Digital flexible ureteroscopes	CMOS (complementary metal-oxide semiconductor) chip at the tip; LED (light-emitting diode) light	Sharper image that does not degrade; tip larger (~8.5 Fr); no more durable
Single-use scopes (LithoVue, 2016)	7.7 Fr tip, ≥270° deflection, new each case	No repair or reprocessing; cheaper below ~99 cases a year; for lower pole and difficult cases
Thulium fibre laser (about 1940 nm)	Long pulse, very high frequency	Fine dust, minimal retropulsion; faster than Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) at low settings
Pulse-modulated (Moses) Ho:YAG	Pulse shaped through a vapour bubble	Less retropulsion, faster dusting
Access sheaths (12/14 Fr)	Outflow channel	Lower intrarenal pressure and temperature; severe ureteric injury in ~1 in 8
Suction sheaths and FANS (flexible and navigable suction access sheath)	Vacuum at the sheath; flexible tip follows the scope into calyces	Aspirates dust; lower pressure; better stone-free rates in early studies
Intrarenal pressure monitoring	Sensor in the wire or scope (LithoVue Elite)	High pressure causes backflow, SIRS (systemic inflammatory response syndrome) and sepsis; safe limits not yet defined
Robotic flexible ureteroscopy (Avicenna Roboflex)	Seated console controls the scope	Ergonomics, less radiation to surgeon

- **Laser safety**: tissue damage above 43 °C, so keep power ≤10 W in the ureter and ≤25 W in the kidney, with continuous cool irrigation.
- **Also know:** dusting versus fragment-and-basket; nitinol tipless baskets; anti-retropulsion devices; Monarch robotic platform for PCNL (percutaneous nephrolithotomy) access.

Diagram to draw: 24-h urine → abnormality → drug flowchart.

Key points to remember:

- Urine ≥2.5 L/day; Borghi 1996: 12% vs 27% recurrence.
- Thiazide plus potassium citrate for hypercalciuria; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

IV-3a. Saturation biopsies of prostate

Type: SAQ (10 marks)

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often about 20–30). It samples the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone. The 12-core scheme (adds apical and far-lateral cores) is today's standard.
- Saturation aims to cut sampling error, mainly for **anterior and apical cancers** (anterior-only cancer 6–20%).

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation and **focal therapy** mapping (shows multifocality).

Templates and technique

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine transrectally).

Results

- **Initial biopsy**: 12 to 18–21 cores did not raise detection.
- **Repeat setting**: early series gained; a large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS (transrectal ultrasound) biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention** than standard biopsy; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** found more significant cancers (38% vs 26%) and 28% avoided biopsy.
- **EAU (European Association of Urology) 2023** prefers the transperineal route. Saturation remains for MRI-negative high suspicion and focal therapy mapping.

Diagram to draw: transperineal grid over the prostate with sampling sectors.

Key points to remember:

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

IV-3b. Testis sparing surgeries for CA testis

Type: SAQ (10 marks)

Definition and rationale

- **Partial orchidectomy**: inguinal excision of the tumour, preserving the rest of the testis. The standard remains **radical inguinal orchidectomy**; sparing aims to keep androgen production and avoid lifelong testosterone.

Indications

Suspected germ cell tumour (GCT), all of:

- Synchronous bilateral tumours or tumour in a **solitary testis**.
- Tumour **≤2 cm** (up to 2–3 cm, under 30% of testicular volume), preferably polar.
- Adequate preoperative testosterone.
- Not with a normal contralateral testis.

Suspected benign or indeterminate lesions:

- Small impalpable lesion (<10 mm) with normal markers: most benign (cyst, infarct, Leydig nodule), but 20–50% are small seminomas. Lesions <5 mm are rarely malignant.
- **Prepubertal**: teratoma and epidermoid cyst with normal AFP (alpha-fetoprotein): partial orchidectomy is the standard.

Preoperative work-up

- Ultrasound of both testes; AFP, beta-hCG (human chorionic gonadotropin), LDH (lactate dehydrogenase); testosterone, LH (luteinising hormone); **sperm banking**; consent for radical orchidectomy.

Technique

1. Inguinal incision; early cord control with a soft clamp.
2. Deliver the testis; isolate with towels.
3. **Intraoperative ultrasound** localises the lesion.
4. Cold ischaemia optional if resection <30 minutes.
5. Incise the tunica parallel to the segmental vessels; excise the mass with a rim of normal tubules.
6. **Frozen section** (benign versus malignant) and biopsies of adjacent parenchyma for GCNIS (germ cell neoplasia in situ).
7. Close the tunica; fix the testis in the scrotum.

Management of GCNIS

- **Radiotherapy about 18–20 Gy** to the remnant: prevents invasive GCT, keeps Leydig function, but causes sterility; up to 40% later need testosterone.
- German Testicular Cancer Study Group: no local recurrences with radiotherapy; most men who omitted it recurred.
- Radiotherapy may be deferred for paternity under close follow-up. Local recurrence → completion orchidectomy.

Complications and follow-up

- Haematoma, atrophy, hypogonadism, local recurrence. Follow with self-examination, ultrasound, markers and testosterone.

Diagram to draw: testis with polar tumour excised with a rim of normal tissue.

Key points to remember:

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

IV-3c. Sacral neuromodulation.

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator).
- It acts on **afferent** nerves, not motor nerves. In OAB (overactive bladder) it damps the PMC (pontine micturition centre). In retention it restores filling sensation, so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

Indications (third line)

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997; urgency-frequency 1999	Bladder pain, pelvic pain
Non-obstructive retention (e.g. Fowler's syndrome) 1999; faecal incontinence 2011	Neurogenic bladder

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography); responds well. Retention needs some detrusor function.

Technique

Test phase	PNE (percutaneous nerve evaluation)	Staged tined lead
Lead	Temporary, office	Permanent, theatre
Duration	5–7 days	Up to 4 weeks
Use	OAB	More reliable; retention

- Prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot rotation
S3	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows only	None

- **Implant if ≥50% improvement** in a diary parameter (retention: 50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

Outcomes

- **InSite**: 61% vs 42% success over medical therapy at 6 months.
- **ROSETTA**: onabotulinumtoxinA 200 U slightly better for UUI, but more UTI (urinary tract infection) and self-catheterisation.

Complications and contraindications

- Pain, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costly.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

Recent advances

- Rechargeable devices (≥15 years); MRI (magnetic resonance imaging)-conditional systems; PTNS (percutaneous tibial nerve stimulation) as an alternative.

Diagram to draw: lead in S3 foramen, IPG in the buttock.

Key points to remember:

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves ≥50%.

IV-3d. Video Urodynamics

Type: SAQ (10 marks)

Definition

- **VUDS (videourodynamics)**: multichannel urodynamics (pressure–flow, with or without EMG (electromyography)) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.

Set-up

- Vesical and rectal catheters (Pdet (detrusor pressure) = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table. Position for the question: **SUI (stress urinary incontinence) standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

What fluoroscopy adds

Situation	VUDS finding
Neurogenic bladder	**Gold standard** (EAU); DESD (detrusor–external sphincter dyssynergia) and DISD (detrusor–internal sphincter dyssynergia); trabeculation, diverticula
VUR (vesicoureteric reflux)	Volume, pressure and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe
Bladder neck	Only way to see it; **PBNO** (primary bladder neck obstruction) in young men and women
Female obstruction	**Nitti criteria**: radiographic narrowing between bladder neck and meatus during voiding
Dysfunctional voiding	"Spinning-top" urethra at the external sphincter
SUI	Leakage with ALPP (abdominal leak point pressure); open bladder neck at rest suggests ISD (intrinsic sphincter deficiency)
Male obstruction	Localises prostatic obstruction, bladder neck contracture or stricture

PBNO types: I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

Indications

- Known or suspected NLUTD (neurogenic lower urinary tract dysfunction); unexplained retention in women; young men with LUTS (lower urinary tract symptoms); complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children**: PUV (posterior urethral valves), spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

Urodynamic risk factors seen

- Impaired compliance; DESD; DISD; high-pressure DO (detrusor overactivity); DLPP (detrusor leak point pressure) >40 cm H_2O; poor emptying with high storage pressure.

Limitations

- Radiation, cost, space, availability, contrast allergy. AUA (American Urological Association) guideline: lowest dose that answers the question.

Diagram to draw: VUDS trace (Pves, Pabd, Pdet, flow, EMG) beside a voiding cystogram image.

Key points to remember:

- VUDS = urodynamics + fluoroscopy with contrast.
- Only test for bladder neck dysfunction and DISD.
- Reflux pop-off makes compliance look falsely good.
- DLPP >40 cm H_2O endangers upper tracts.

IV-3e. Tubeless PCNL

Type: SAQ (10 marks)

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant perforation of the collecting system
Supracostal access, bilateral cases (shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent through the urethra or a string.
3. Withdraw the sheath under vision; **safety wire is removed last**.
4. Watch the tract; apply pressure. Fibrin glue in the tract did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and operative time.
- Less pain than large tubes (**>16 Fr**), but **no benefit against small 8–9 Fr tubes**.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents are removed before discharge without cystoscopy.

Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

Current position

- Choose tubeless, small tube or stent case by case (AUA (American Urological Association)). Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

Key points to remember:

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

IV-3f. Vesicostomy.

Type: SAQ (10 marks)

Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

Alternatives

Option	Use
Cutaneous ureterostomy or pyelostomy	Rare; persistent upper tract obstruction
Ileovesicostomy	Unable to catheterise; 10–15 cm ileum

Option	Use
Appendicovesicostomy (Mitrofanoff, 1980)	Continent catheterisable channel

Diagram to draw: sagittal view of bladder dome sutured to fascia and skin.

Key points to remember:

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.
- Stenosis and prolapse are the main complications.

Rapid Revision Sheet – Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

IV-1. Orthotopic versus Non Orthotopic bladder. (LAQ)

- Neobladder needs eGFR >35–40, a tumour-free urethral margin and ability to self-catheterise.
- Day continence 80–90%; night incontinence 20–50%.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- No diversion is proven best for quality of life; counsel all options.

IV-2. Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS. (LAQ)

- Urine ≥2.5 L/day; Borghi 1996: 12% vs 27% recurrence.
- Thiazide plus potassium citrate for hypercalciuria; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

IV-3a. Saturation biopsies of prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

IV-3b. Testis sparing surgeries for CA testis (SAQ)

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

IV-3c. Sacral neuromodulation. (SAQ)

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
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- Tubeless = stent only; totally tubeless = neither.
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IV-3f. Vesicostomy. (SAQ)

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